

Vanshika Bhatia

[Portfolio](#) | [LinkedIn](#) | vbhati05@student.ubc.ca | +1 (778) 956-0402

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science, Manufacturing Engineering, Minor in Commerce

Expected: May 2028

Relevant Coursework: Manufacturing Processes, Production Systems Management, Engineering Materials, Mechanics of Materials, Engineering Measurements & Instrumentation, Statistics, Control of Mechatronic Systems

WORK EXPERIENCE

Mechanical Engineering Intern - Battery Stack Team | Invinity Energy Systems

Jun 2026 – Dec 2026

Onshape, GD&T, DFM, 3D Printing, Pneumatics, Vacuum Systems, ECOS, Excel

- Designed a pneumatic vacuum-assisted flip fixture using Onshape targeting 80% faster felt placement, saving 2.8 operator-hours/day across 200 cells and projected to reduce felt shifting defects by 75%
- Built a pulley-counterweight system, cutting effective plate load by 80% for controlled flipping and future actuator integration
- Validated fixture fit with 3D-printed prototypes and released production-ready GD&T drawings and BOM through ECOS
- Characterized temperature-dependent stack creep using heated foam-insulation testing, risk assessment, and Excel analysis with time-temperature superposition

Quality and Continuous Improvement Engineering Intern | Phase Technology

Sep 2025 – Dec 2025

FMEA, RCA, Lean Manufacturing, VSM, 5S, Kanban, Incoming Inspection, Nonconformance, Change Control, Quality Documentation

- Targeted 1.5 operator-hours saved daily by mapping process waste through VSM and implementing 5S and Kanban Systems
- Inspected 30+ machined and PCB parts weekly against drawings, containing defects through dimensional and leak testing
- Investigated recurring leak failures with R&D using FMEA, root cause analysis, and experiments across designs and processes
- Owned 100+ open MRP dispositions in fast-paced production across engineering, production, and suppliers, while managing work instructions and ECOS change control in Excel
- Improved inspection repeatability by establishing calibration and verification controls for test and measurement equipment

PROJECTS & ENGINEERING DESIGN TEAMS

RC Car Redesign, UBC Manufacturing, Design Lead

Sep 2024 - Apr 2025

- Redesigned aluminum wheels in SolidWorks and applied FMEA to identify failure modes and drive design improvements
- Sand-casted lightweight wheels and developed thermoformed car shell and chassis extension for improved stability

Design Team, Electrical SubTeam Member

Sep 2023 - Sep 2025

- Designed and validated a Sabatier reactor control PCB in KiCad and LTSpice, improving switching reliability and manufacturability through optimized MOSFET, zero-cross, and TRIAC circuitry
- Set reactor temperature through PID control, integrating sensor hardware and control software for closed-loop regulation

1U CubeSat, Space Technology & Aeronautical Rocketry, Mechatronics Intern

May 2023 - Aug 2023

- Designed a lightweight 1U CubeSat frame in Fusion 360 using DFM and validated manufacturability through 3D printing
- Co-designed an Eagle PCB preserving 70% of board space for payloads while integrating Arduino Nano control, CST antenna design, and C based communications

SKILLS

CAD & Design: SolidWorks, Onshape, Fusion 360, GD&T, DFMA, Engineering Drawings, BOMs

Manufacturing & Testing: 3D Printing, Injection Molding, Lathe, Milling, Pneumatics, Leak Testing, Tensile Testing

Quality & Process: FMEA, RCA, VSM, 5S, Kanban, Nonconformance Management, Change Control, ECOS, Excel